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# 第五章 导数相关综合试卷

## 一、 选择题（Multiple Choice Questions）

1. 函数 $y=\sqrt{x}$ 的导数是 ( ). What is the derivative of the function $y=\sqrt{x}$? ( )

   A. $y=\sqrt{x}$

   B. $y=\frac{1}{\sqrt{x}}$

   C. $y=\frac{1}{2\sqrt{x}}$

   D. $y=-\frac{1}{2\sqrt{x}}$

2. 曲线 $y=2^x$ 在点 $P(3,8)$ 处的切线斜率是 ( ). What is the slope of the tangent line of the curve $y=2^x$ at the point $P(3,8)$? ( )

   A. $7\ln 2$

   B. $6\ln 2$

   C. $9\ln 2$

   D. $8\ln 2$

3. 曲线 $y=\ln x$ 在点 $P(1,0)$ 处的切线方程是 ( ). What is the equation of the tangent line of the curve $y=\ln x$ at the point $P(1,0)$? ( )

   A. $y=x-1$

   B. $y=x-e$

   C. $y=ex-1$

   D. $y=ex-e$

4. 若物体的运动方程是 $s=4t^2+\frac{1}{t}$, 则物体在 $t=2$ 时的瞬时速度是 ( ). If the motion equation of an object is $s=4t^2+\frac{1}{t}$, what is the instantaneous

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velocity of the object at $t = 2$? ( )

A. $\frac{33}{2}$

B. $\frac{65}{4}$

C. $\frac{65}{2}$

D. $\frac{63}{4}$

5. 设函数 $f(x) = \sin x$，则 $f'(0)$ 等于 ( ). Let the function $f(x) = \sin x$, then what is $f'(0)$ equal to? ( )

A. 1

B. -1

C. 0

D. None of the above

6. 设函数 $f(x) = \frac{4}{x}$，则 $f'(1)$ 等于 ( ). Let the function $f(x) = \frac{4}{x}$, then what is $f'(1)$ equal to? ( )

A. 1

B. 4

C. -4

D. None of the above

7. 函数 $y = 4(2 - x + 3x^2)^2$ 的导数是 ( ). What is the derivative of the function $y = 4(2 - x + 3x^2)^2$? ( )

A. $8(2 - x + 3x^2)$

B. $2(-1 + 6x)^2$

C. $8(2 - x + 3x^2)(6x - 1)$

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D. $4(2-x+3x^2)(6x-1)$

8. 如果函数 $y=x^4$, 则 $y^{(5)}$ 等于（). If the function $y=x^4$, then what is $y^{(5)}$ equal to? ( )

A. $4x^3$

B. 0

C. $24x$

D. None of the above

9. 函数 $f(x)=x^3-3x-1$ 的三阶导数是（). What is the third derivative of the function $f(x)=x^3-3x-1$? ( )

A. $3x^2-3$

B. $6x$

C. 6

D. 0

10. 函数 $y=\sin x+2$ 的四阶导数是（). What is the fourth derivative of the function $y=\sin x+2$? ( )

A. $\sin x$

B. $\cos x$

C. $\sin x+2$

D. 0

11. 如果 $y=\ln x$，那么 $y^{(4)}$ 等于（). If $y=\ln x$, then what is $y^{(4)}$ equal to? ( )

A. $\frac{1}{x}$

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B. $-\frac{1}{x^2}$

C. $\frac{2}{x^3}$

D. $-\frac{6}{x^4}$

12.如果 $y = y(x)$ 是 $x - y = 0$ 确定的隐函数，$\frac{dy}{dx} = (\ )$. If $y = y(x)$ is an implicit function determined by $x - y = 0$, then $\frac{dy}{dx} = (\ )$

A. -1

B. 0

C. 1

D. None of the above

13.如果 $y = y(x)$ 是 $e^y + x^3 + x = 0$ 确定的隐函数，$\frac{dy}{dx} = (\ )$. If $y = y(x)$ is an implicit function determined by $e^y + x^3 + x = 0$, then $\frac{dy}{dx} = (\ )$

A. $-\frac{3x^2+1}{e^y}$

B. $\frac{3x^2+1}{e^y}$

C. $\frac{3x^2-1}{e^y}$

D. None of the above

14.如果 $y = y(x)$ 是 $x^2 - xy = 0$ 确定的隐函数，$y' = (\ )$. If $y = y(x)$ is an implicit function determined by $x^2 - xy = 0$, then $y' = (\ )$

A. $\frac{x-2y}{x}$

B. $\frac{2x+y}{x}$

C. $\frac{2x-y}{x}$

D. None of the above

15.如果 $y = y(x)$ 是 $\ln y - x = 0$ 确定的隐函数，$y' = (\ )$. If $y = y(x)$ is an

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implicit function determined by $\ln y - x = 0$, then $y' =$ ( )

A. $\frac{1}{y}$

B. $y$

C. $\frac{x}{y}$

D. None of the above

16.如果 $y = y(x)$ 是 $\begin{cases} x = t - 1 \\ y = 2t^3 - 3t^2 + 1 \end{cases}$ 确定的参数函数，$\frac{dy}{dx} =$ ( ). If $y = y(x)$ is a parametric function determined by $\begin{cases} x = t - 1 \\ y = 2t^3 - 3t^2 + 1 \end{cases}$, then $\frac{dy}{dx} =$ ( )

A. $2t^2 - 3t$

B. 1

C. $6t^2 - 6t$

D. None of the above

17.如果 $y = y(x)$ 是 $\begin{cases} x = -\mathrm{sin}t \\ y = 2\mathrm{cos}t \end{cases}$ 确定的参数函数，$\frac{dy}{dx} =$ ( ). If $y = y(x)$ is a parametric function determined by $\begin{cases} x = -\mathrm{sin}t \\ y = 2\mathrm{cos}t \end{cases}$, then $\frac{dy}{dx} =$ ( )

A. $-2\mathrm{tan}t$

B. $2\mathrm{tan}t$

C. $2\mathrm{cot}t$

D. None of the above

18.如果 $y = y(x)$ 是 $\begin{cases} x = \mathrm{ln}t \\ y = e^t \end{cases}$ 确定的参数函数，$y' =$ ( ). If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \mathrm{ln}t \\ y = e^t \end{cases}$, then $y' =$ ( )

A. $te^t$

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B. $-te^t$

C. $\frac{e^t}{t}$

D. None of the above

19.若函数 $f(x)=x^{-2}$，则 $f'(1)$ 等于（ ). If the function $f(x)=x^{-2}$, then what is $f'(1)$ equal to? ( )

A. 0

B. -1

C. -2

D. 1

20.函数 $f(x)=-10$ 的导数是（ ). What is the derivative of the function $f(x)=-10$? ( )

A. 0

B. Negative number

C. Positive number

D. Uncertain

21.若 $f(x)=\sqrt[3]{x}$，则 $f'(1)$ 等于（ ). If $f(x)=\sqrt[3]{x}$, then what is $f'(1)$ equal to? ( )

A. 0

B. $\frac{1}{3}$

C. 1

D. $\frac{3}{2}$

22.设函数 $f(x)=-\cos x$，则 $f'\left(\frac{\pi}{2}\right)$ 等于（ ). Let the function $f(x)=-\cos x$,

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then what is $f'\left(\frac{\pi}{2}\right)$ equal to? ( )

A. 0

B. 1

C. -1

D. None of the above

23.$y = x^n$ 在 $x = 1$ 处切线方程为 $y = -4x + 5$，则 $n$ 的值为（ ）. The tangent line equation of $y = x^n$ at $x = 1$ is $y = -4x + 5$, then what is the value of $n$? ( )

A. 4

B. -4

C. 1

D. -1

24.如果 $y = x\ln x$，那么 $y^{(4)}$ 等于（ ）. If $y = x\ln x$, then what is $y^{(4)}$ equal to? ( )

A. $\ln x + 1$

B. $\frac{1}{x}$

C. $-\frac{1}{x^2}$

D. $\frac{2}{x^3}$

25.如果 $y = y(x)$ 是 $y^2 - xy = 0$ 确定的隐函数，$\frac{dy}{dx} =$ ( ). If $y = y(x)$ is an implicit function determined by $y^2 - xy = 0$, then $\frac{dy}{dx} =$ ( )

A. $\frac{2y+x}{y}$

B. $\frac{2y-x}{y}$

C. $\frac{y}{2y-x}$

D. $\frac{y}{2y+x}$

## 二、 填空题（Fill-in-the-Blank Questions）

1. 函数 $y=\sqrt{x}\cdot 3^x$ 的导数是________.

   What is the derivative of the function $y=\sqrt{x}\cdot 3^x$? ________

2. 曲线 $y=\frac{1}{x^2}$ 在点 $A\left(2,\frac{1}{4}\right)$ 处的切线斜率是________；切线方程是________. What is the slope of the tangent line of the curve $y=\frac{1}{x^2}$ at the point $A\left(2,\frac{1}{4}\right)$? ________; What is the equation of the tangent line? ________

3. 若物体的运动方程是 $s=4t^2+3$，则物体在 $t=1$ 时的瞬时速度是________. If the motion equation of an object is $s=4t^2+3$, what is the instantaneous velocity of the object at $t=1$? ________

4. 函数 $y=-\frac{1}{2}x^{\frac{1}{2}}$，则 $y'=$________.

   Given the function $y=-\frac{1}{2}x^{\frac{1}{2}}$, then $y'=$ ________

5. 曲线 $y=\frac{1}{3}x^3$ 在 $x=1$ 处切线的倾斜角为________.

   What is the inclination angle of the tangent line of the curve $y=\frac{1}{3}x^3$ at $x=1$? ________

6. $y=x^3+\sqrt{x}$ 的导数为________.

   What is the derivative of $y=x^3+\sqrt{x}$? ________

7. $y=x\sin 2x$ 的导数为________.

   What is the derivative of $y=x\sin 2x$? ________

8. $y=-\tan x$ 的导数为________.

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What is the derivative of $y = -\tan x$? ________

9. $y = -2x^3 + \sqrt{x}$ 的三阶导数为________.

What is the third derivative of $y = -2x^3 + \sqrt{x}$? ________

10. $y = x\cos x$ 的四阶导数为________.

What is the fourth derivative of $y = x\cos x$? ________

11. $y = -\tan x$ 的二阶导数为________.

What is the second derivative of $y = -\tan x$? ________

12. $y = \ln(1 + x)$ 的四阶导数为________.

What is the fourth derivative of $y = \ln(1 + x)$? ________

13. 如果 $y = y(x)$ 是 $xy - y^2 = 0$ 确定的隐函数，$\frac{dy}{dx} =$ ________.

If $y = y(x)$ is an implicit function determined by $xy - y^2 = 0$, then $\frac{dy}{dx} =$ ________

14. 如果 $y = y(x)$ 是 $\sin x - y = 0$ 确定的隐函数，$\frac{dy}{dx} =$ ________.

If $y = y(x)$ is an implicit function determined by $\sin x - y = 0$, then $\frac{dy}{dx} =$ ________

15. 如果 $y = y(x)$ 是 $x^3 - xy + y^2 = 0$ 确定的隐函数，$\frac{dy}{dx} =$ ________.

If $y = y(x)$ is an implicit function determined by $x^3 - xy + y^2 = 0$, then $\frac{dy}{dx} =$ ________

16. 如果 $y = y(x)$ 是 $\sin(x - y) + x = y$ 确定的隐函数，$\frac{dy}{dx} =$ ________.

If $y = y(x)$ is an implicit function determined by $\sin(x - y) + x = y$, then $\frac{dy}{dx} =$ ________

17. 如果 $y = y(x)$ 是 $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$ 确定的参数函数，$\frac{dy}{dx} =$ ________.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$, then $\frac{dy}{dx} =$ ________

18.如果 $y = y(x)$ 是 $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$ 确定的参数函数，$\frac{d^2y}{dx^2} =$ ________.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \cos 2t \\ y = \sin 6t \end{cases}$, then $\frac{d^2y}{dx^2} =$ ________

19.如果 $y = y(x)$ 是 $\begin{cases} x = e^{-t} \\ y = \ln 2t \end{cases}$ 确定的参数函数，$\frac{dy}{dx} =$ ________.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = e^{-t} \\ y = \ln 2t \end{cases}$, then $\frac{dy}{dx} =$ ________

20.曲线 $y = x^n$ 在 $x = 2$ 处的导数为 12，则 $n$ 等于________.

The derivative of the curve $y = x^n$ at $x = 2$ is 12, then what is $n$ equal to? ________

21.函数 $y = \frac{e^x}{x}$，则 $y' =$ ________.

Given the function $y = \frac{e^x}{x}$, then $y' =$ ________

22.函数 $y = x\ln x$，则 $y' =$ ________.

Given the function $y = x\ln x$, then $y' =$ ________

23.$y = 1 - \frac{1}{x^2}$ 的导数为________.

What is the derivative of $y = 1 - \frac{1}{x^2}$? ________

24.$y = (x+1)^2(x-1)$ 的导数为________；在 $x = 1$ 处的导数为________.

What is the derivative of $y = (x+1)^2(x-1)$? ________; What is the derivative at $x = 1$? ________

25.如果 $y = \sin^2 x$，那么 $y^{(4)} =$ ________.

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If $y = \sin^2 x$, then $y^{(4)} =$ ________

26.如果 $y = y(x)$ 是 $e^y \ln x - xy = 0$ 确定的隐函数，$\frac{dy}{dx} =$ ________.

If $y = y(x)$ is an implicit function determined by $e^y \ln x - xy = 0$, then $\frac{dy}{dx} =$ ________

## 三、 计算题（Calculation Questions）

1. 求函数 $y = 2 - 3x^3$ 的导数.

   Find the derivative of the function $y = 2 - 3x^3$.

2. 求曲线 $y = \frac{2}{x} - \frac{1}{x^2}$ 在点 $A\left(2, \frac{3}{4}\right)$ 处的切线斜率和切线方程.

   Find the slope and the equation of the tangent line of the curve $y = \frac{2}{x} - \frac{1}{x^2}$ at the point $A\left(2, \frac{3}{4}\right)$.

3. 求下列函数的导数. Find the derivatives of the following functions.

   （1）$y = \frac{\sin x}{x}$

   （2）$y = \cos^2 x - 2x + 1$

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4. 如果 $y=\sin^2 x-3x^2+2$，求 $y'''$.

If $y=\sin^2 x-3x^2+2$, find $y'''$.

5. 如果 $y=\sin 3x+5x^4$，求 $y^{(4)}$.

If $y=\sin 3x+5x^4$, find $y^{(4)}$.

6. 如果 $y=y(x)$ 是 $y\ln x-xy^2=0$ 确定的隐函数，求 $\frac{dy}{dx}$.

If $y=y(x)$ is an implicit function determined by $y\ln x-xy^2=0$, find $\frac{dy}{dx}$.

7. 如果 $y=y(x)$ 是 $y^4-2x^2y+3x-5=0$ 确定的隐函数，求 $y'$ 和 $y'|_{x=0}$.

If $y=y(x)$ is an implicit function determined by $y^4-2x^2y+3x-5=0$, find $y'$ and $y'|_{x=0}$.

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8. 如果 $y = y(x)$ 是 $\begin{cases} x = -t + 5 \\ y = -\frac{2}{3}t^3 + t \end{cases}$ 确定的参数函数，求 $\frac{dy}{dx}$ 和 $\frac{d^2y}{dx^2}$.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = -t + 5 \\ y = -\frac{2}{3}t^3 + t \end{cases}$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

9. 如果 $y = y(x)$ 是 $\begin{cases} x = \ln(3t + 2) \\ y = e^{2t} \end{cases}$ 确定的参数函数，求 $y'|_{t=0}$ 和 $y''|_{t=0}$.

If $y = y(x)$ is a parametric function determined by $\begin{cases} x = \ln(3t + 2) \\ y = e^{2t} \end{cases}$, find $y'|_{t=0}$ and $y''|_{t=0}$.

10. 若物体的运动方程是 $s = 8 - 2t + 3t^2$，则物体在 $t = 2$ 时的瞬时速度是多少?

If the motion equation of an object is $s = 8 - 2t + 3t^2$, what is the instantaneous velocity of the object at $t = 2$?

11.求指数函数 $y=a^x$（$a>0$ 且 $a\neq 1$）的导数.

Find the derivative of the exponential function $y=a^x$ ($a>0$ and $a\neq 1$).

12.求曲线 $y=-\sin x$ 在 $x=\pi$ 处的切线方程.

Find the equation of the tangent line of the curve $y=-\sin x$ at $x=\pi$.

13.如果 $y=y(x)$ 是 $\begin{cases}x=2e^{-3t}\\y=\ln 2t\end{cases}$ 确定的参数函数，求 $y'|_{t=1}$ 和 $y''|_{t=1}$.

If $y=y(x)$ is a parametric function determined by $\begin{cases}x=2e^{-3t}\\y=\ln 2t\end{cases}$, find $y'|_{t=1}$ and $y''|_{t=1}$.

14.求下列函数的导数. Find the derivatives of the following functions.

（1）$y=\sqrt{x}$

（2）$y=2\cos x$

（3）$y=3x^3$